

SELF - ATTENTION

- LET'S SUPPOSE WE ARE GIVEN A SEQUENCE OF TOKEN $(e_1, \dots, e_n)$ AND AN INITIAL DESCRIPTION OF THEM $(v_1, \dots, v_n)$
- **GOAL:** WE ARE INTERESTED IN ENRICHING THE TOKEN DESCRIPTOR ADDING MORE CONTEXT

EG.

BANK OF THE RIVER



v_1

v_2

v_3

v_4



} - TOKEN $t_1 \dots t_n$

} - INITIAL DESCRIPTORS

How ?



ATTENTION



ϕ_1

ϕ_2

ϕ_3

ϕ_4

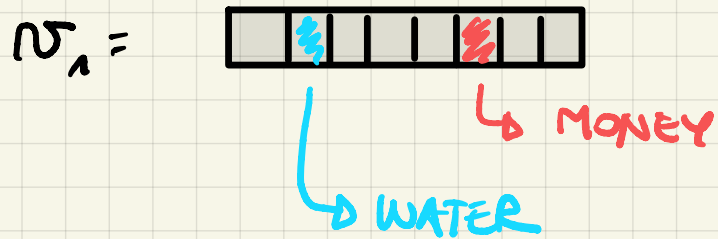
} - NEW DESCRIPTORS WITH MORE CONTEXT

[Applications DOMAIN:

- Language
- Video
- Any Sequence]

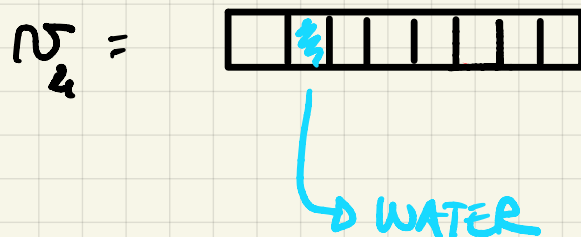
INTUITION:

The descriptor of the token "BANK", should includes both interpretation of BANK linked to money AND water:



If we know only this token with no context, we cannot choose between these two interpretations. It's AMBIGUOUS!

BUT ... Knowing in the same sentence there is the token "RIVER" with a descriptor:



We can reinforce the feature "WATER" and filter the others!

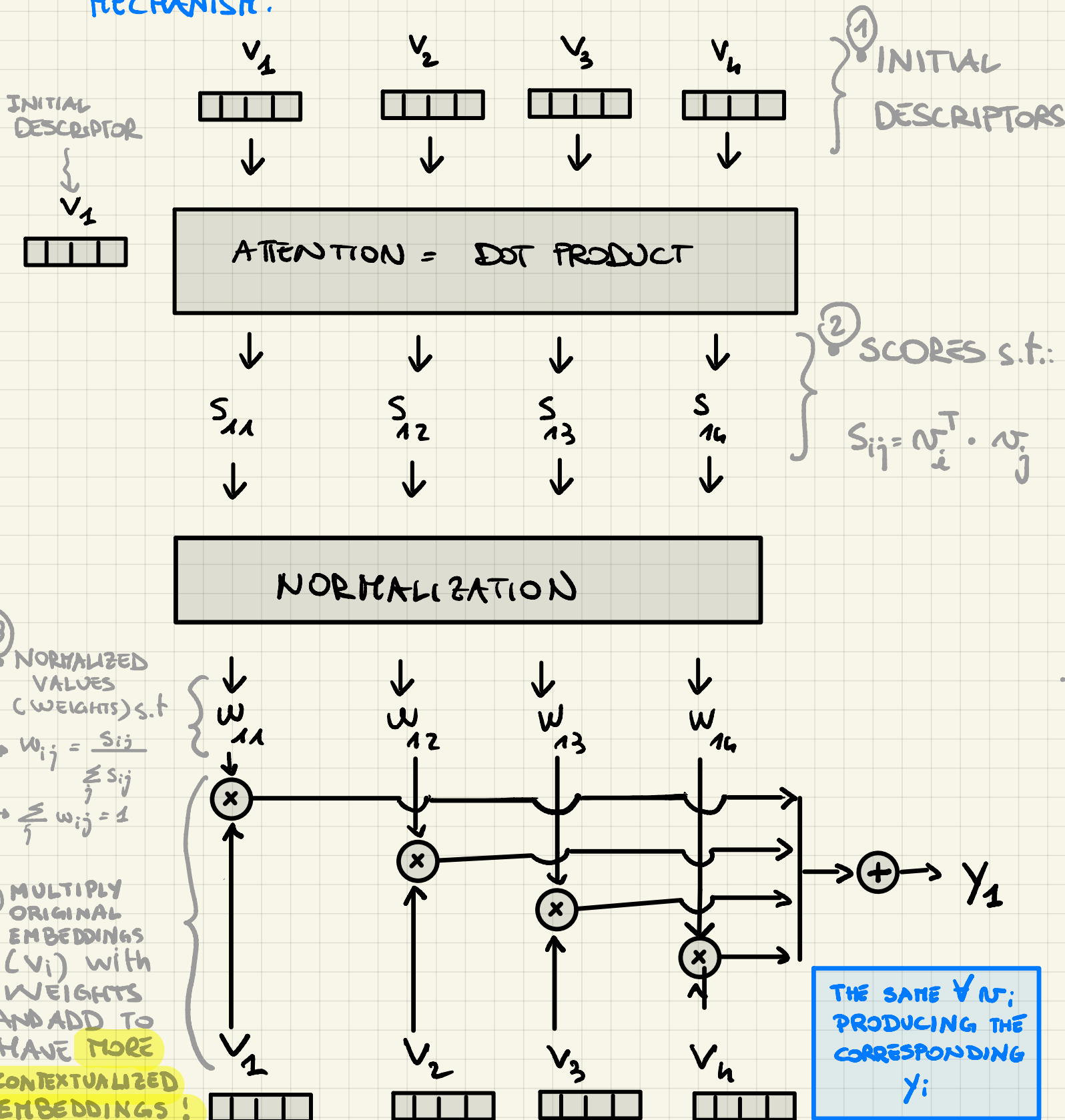
HOW DO WE IMPLEMENT "ATTENTION"?

- LET'S SUPPOSE WE WANT TO ADD CONTEXT

TO THE DESCRIPTOR " V_1 "

- LET'S USE DOT PRODUCT AS ATTENTION

MECHANISM:

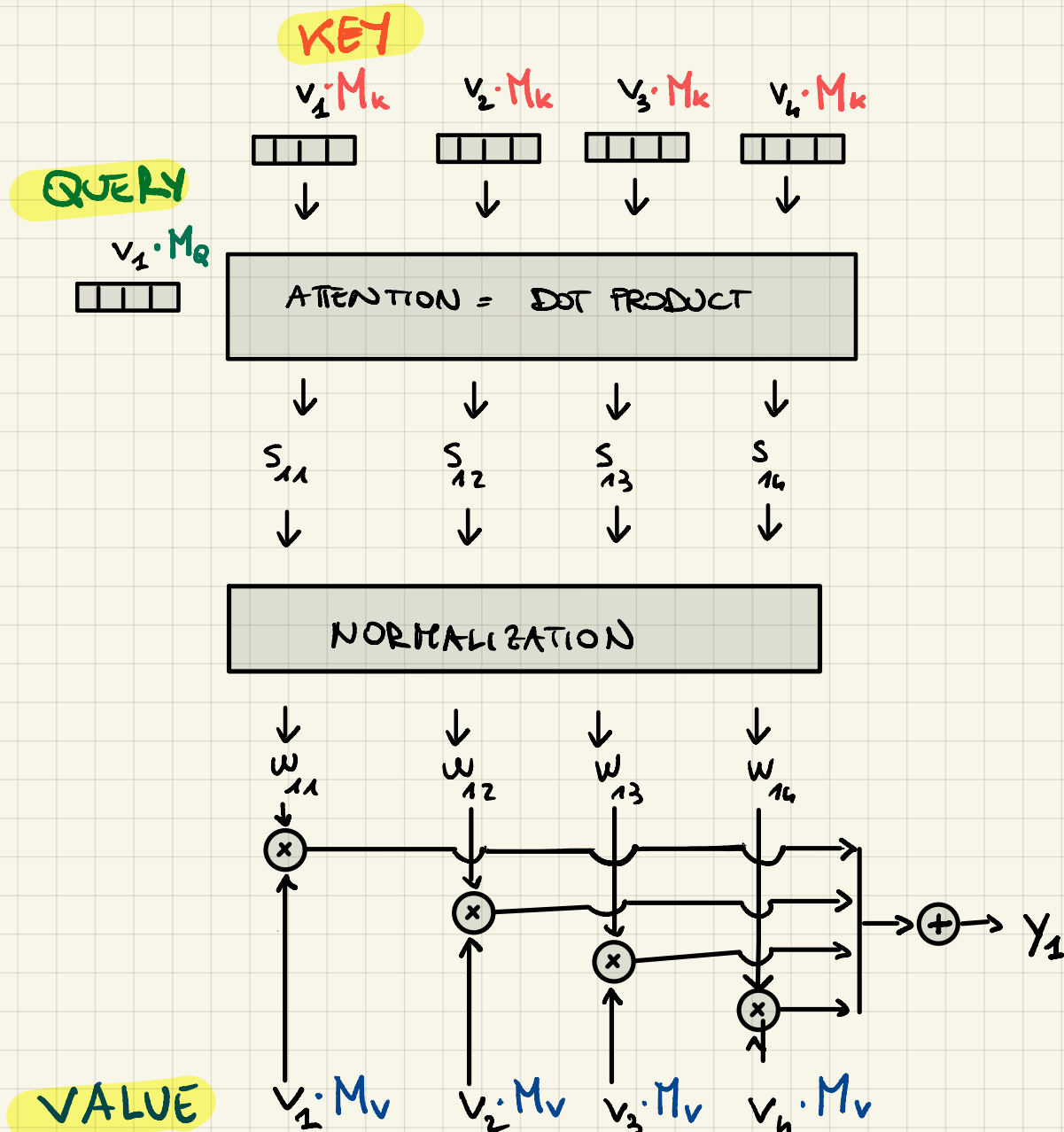


- LET'S ADD TRAINABLE WEIGHTS "M"
(THIS SHOULD ADD FLEXIBILITY):

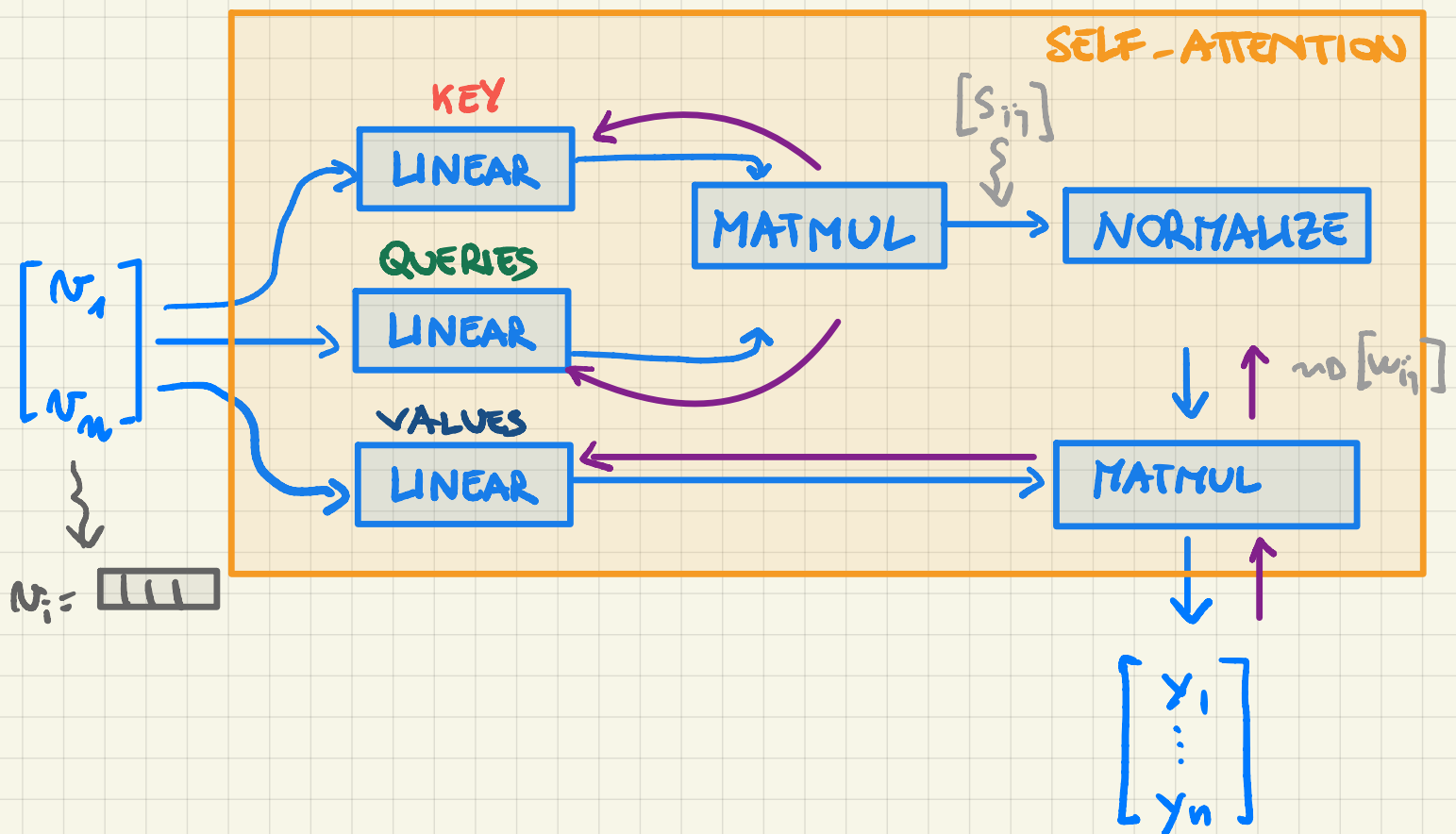
$$N^* = N \cdot M, \text{ s.t. } |N| = |N^*| = 1 \times K$$

and $|M| = K \times K$

- OSS: ORIGINAL DESCRIPTOR IS USED 3 TIMES.
⇒ LET'S ADOPT 3 DIFFERENT MATRICES:



- OSS: WE CAN REDRAW THIS AS A NN:



- WE HAVE DEFINED A NEW BLOCK.

- WE CAN STACK SEVERAL OF THEM:



- AND... WE CAN APPLY BACKPROPAGATION TO LEARN THE SELF ATTENTION MATRICES M_Q, M_K, M_V

MULTI SELF-ATTENTION

eg

I gave my dog Charlie some food.

A diagram illustrating attention weights for the word "gave". Three curved arrows originate from the word "gave" and point to the words "I", "dog", and "some" respectively. The arrows are labeled with circled numbers 1, 2, and 3, indicating different attention heads or weights.

?

⇒ TO WHAT SHOULD WE PAY ATTENTION FOR ENRICHING "GAVE"?

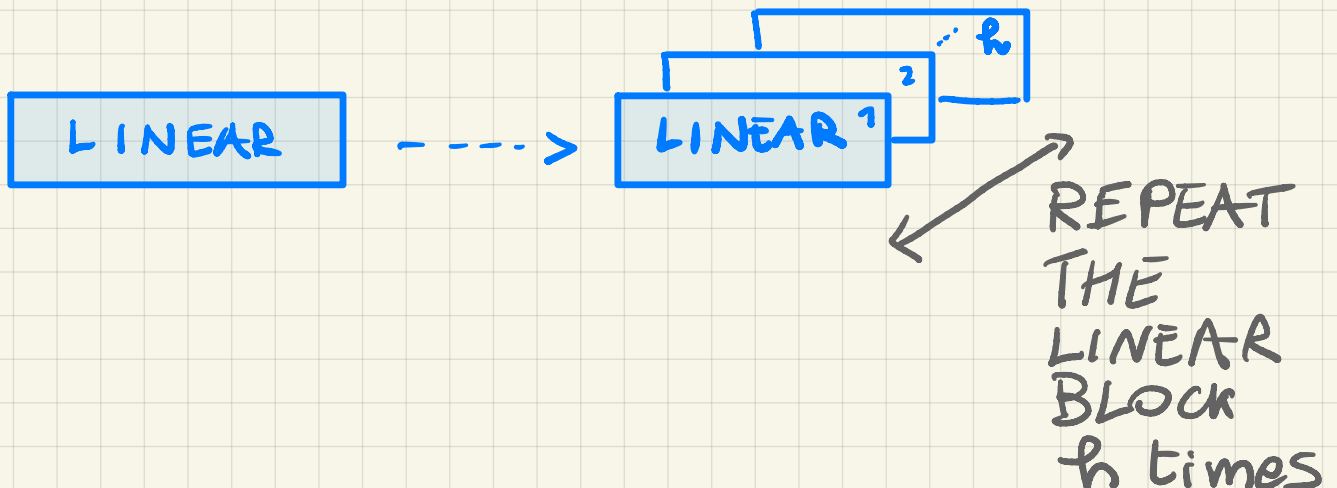
IT COULD BE EITHER 1, 2 OR 3 ...



SOLUTION: MULTI SELF-ATTENTION,
EACH ONE "SPECIALIZED"
ON A PECULIAR FEATURE

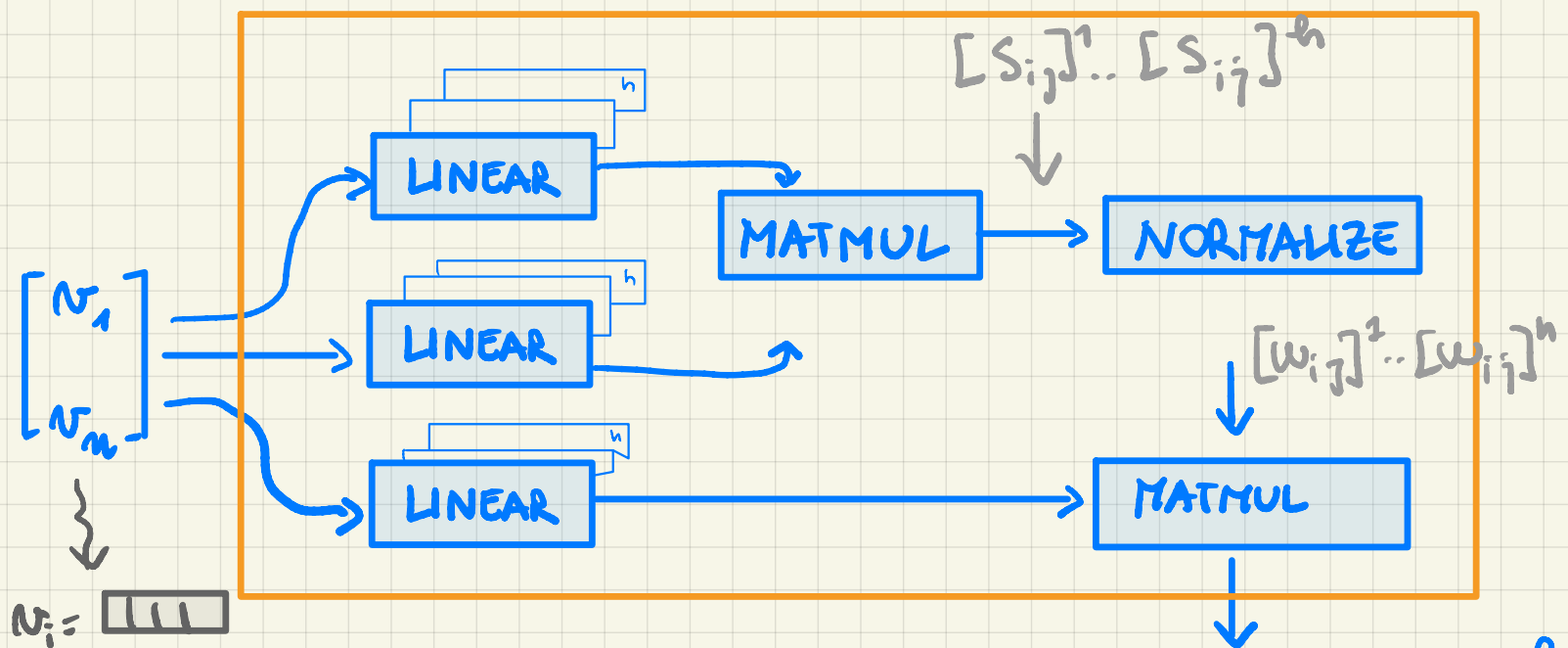
OSS: THEY SHOULD WORK IN PARALLEL

≈ THEY SHOULD NOT SHARE
PARAMETERS



- WHAT WILL IT CHANGE?

MULTI-HEAD ATTENTION



⇒ It will produce h output

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}^1 \dots \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}^i \dots \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}^h$$

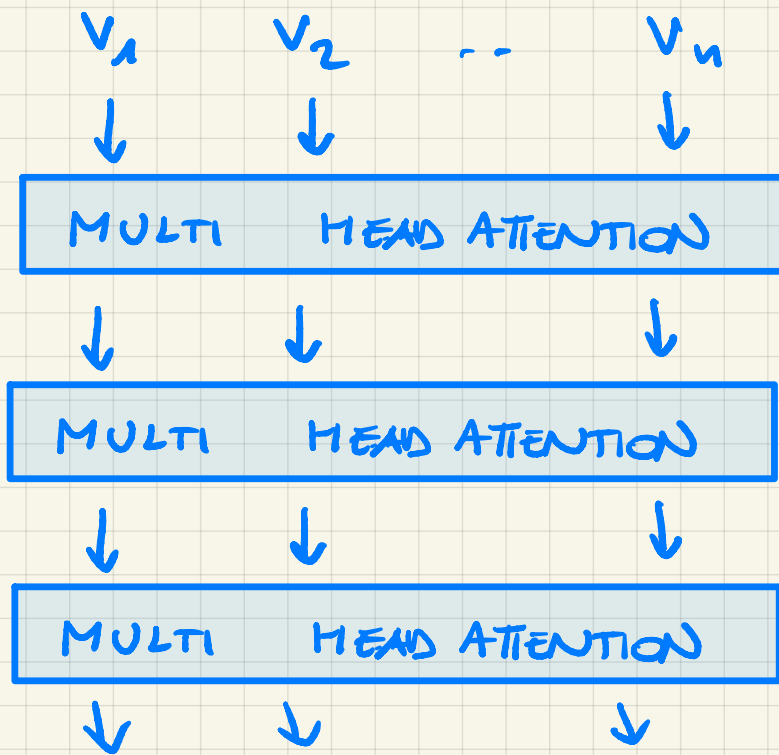
CONCAT + DENSE

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}^*$$

⇒ ADD A NEW
MODULE TO
- CONCATENATE
AND
- MERGE :

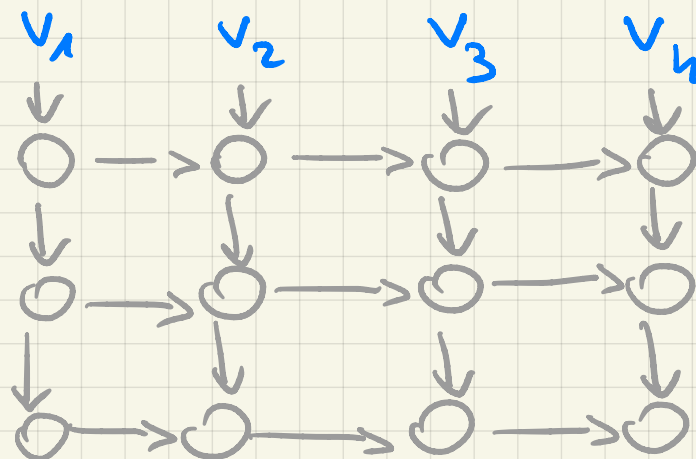
MULTI

MULTI-HEAD ATTENTION



THIS IS THE CORE OF TRANSFORMERS

IT SOLVES 2 PROBLEMS OF THE RECURRENT STRUCTURE:



2 PROBLEMS:

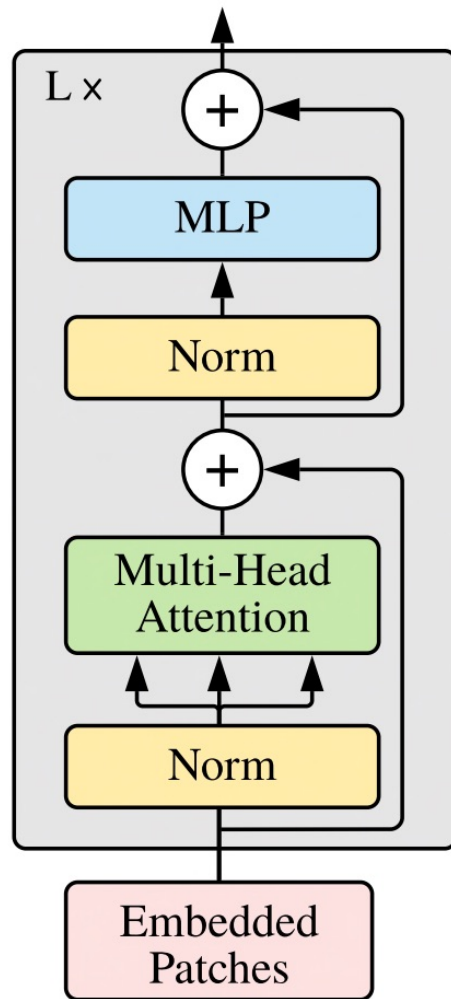
- VANISHING GRADIENT
- NOT PARALLEL

//

TRANSFORMERS

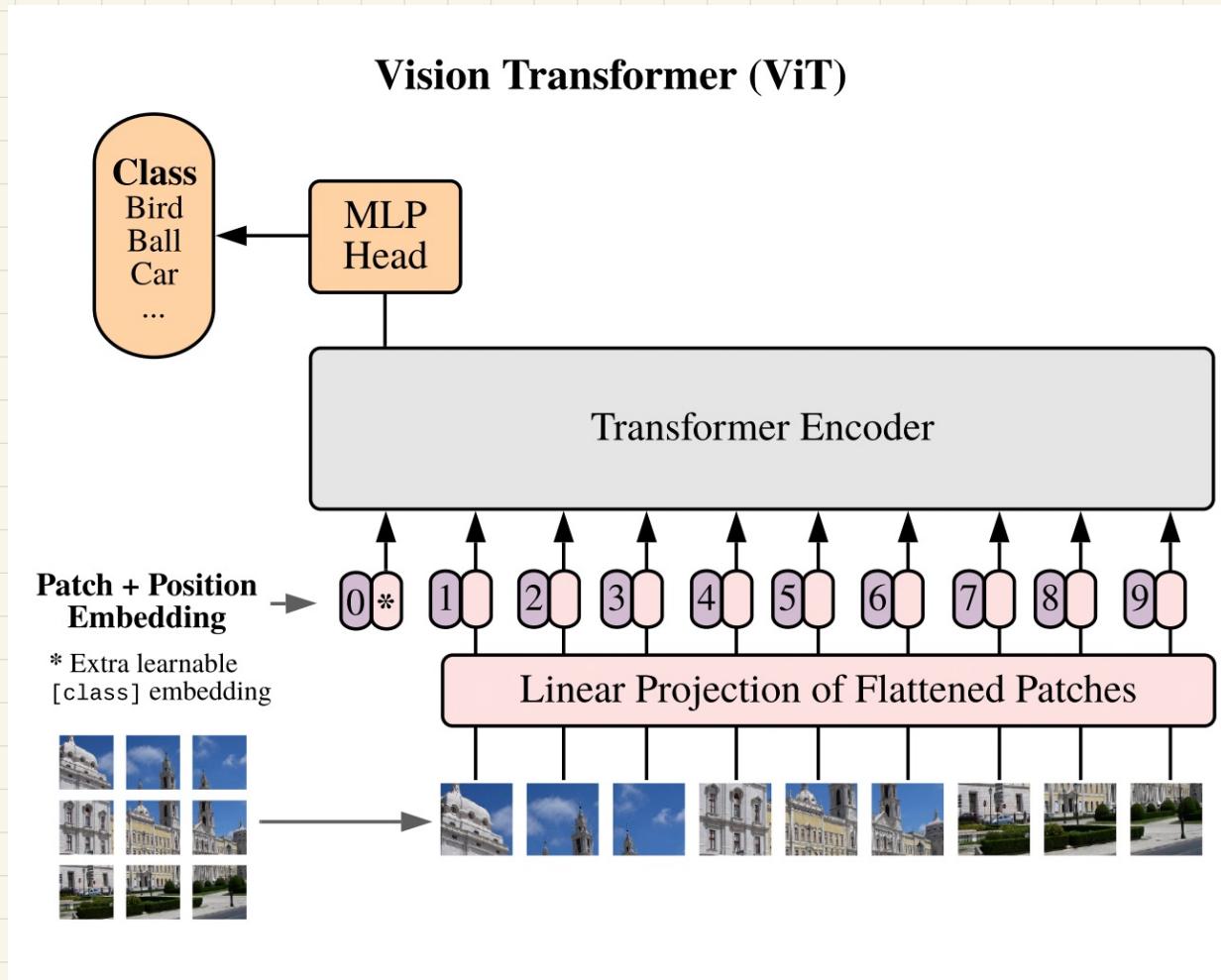
Attention is All you need.

Transformer Encoder



AN IMAGE IS WORTH 16×16 WORDS:

TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE



Is SPACE - TIME ATTENTION All YOU NEED FOR VIDEO UNDERSTANDING?

